Next-Generation Synthetic Face Data with Diffusion Models

Maria Salvador[1]; Rafael Mamede [2,3]; Ana F. Sequeira [2,3]; Hélder Oliveira[2,1]

Faculty of Sciences, University of Porto, Porto, Portugal[1];

INESC TEC, Porto, Portugal[2]; Faculty of Engineering University of Porto, Porto, Portugal[3]

1. INTRODUCTION

PROBLEM STATEMENT

- State-of-the-art deep face recognition (FR) models rely on **large-scale datasets**.
- **Problems:** **regulation changes** [1][2]; **critical data imbalances** (e.g. limited variations, under-representation of protected demographic characteristics).

SYNTHETIC FACE DATA FOR FR TASKS

- Synthetic face datasets for FR in different use-cases [3].
- **Large intra-class diversity** (differences within the same identity) **and large cross-class discrimination** (variability between identities so that the model can generalize well to unseen faces).

DIFFUSION MODELS

- Generative AI models inspired by thermodynamics modeling [4] (Figure 1).
- **Latent Diffusion**: improvement by performing the modeling on a latent space [5].
- **SOTA in image generation**.

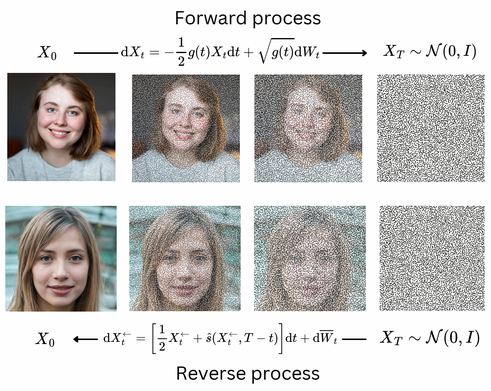

Figure 1. Diffusion Models: Schematic

2. METHODOLOGY

AIM OF THIS WORK

- **Generate a balanced synthetic face dataset** for FR using a **Diffusion Model** based approach.
- **Evaluate** the generated synthetic data in both **photorealistic image quality** and **utility for training FR.** **Does data quality affect its utility?**

DATASET CHOICE

- **BUPT-BalancedFace dataset**, introduced by Wang et al. [6] in 2020, which is divided in **four different ethnicities**, African, Asian, Caucasian and Indian, each containing 7000 identities and a total of ~1.25M images.

MODELS AND METHODS

BASED ON

- **IDiff-Face**, a novel face data generation approach based on conditional LDMs proposed by Boutros et. al. [7] in 2023 (Figure 2).

OUR APPROACH

1. Train the denoising UNet of **four separate IDiff-Face models**, **one per ethnicity.**
2. Use the **knowledge of the models' noise predictions** at different steps of the denoising process to **train a general model -** repurposing **knowledge distillation** as a mechanism for ensemble knowledge fusion.

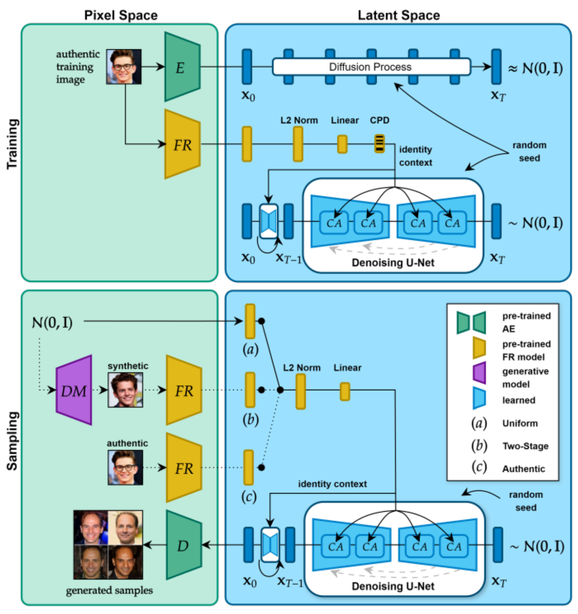

Figure 2. Overview of the proposed IDiff-Face. From the original paper [6]. Contextual Partial Dropout (CPD) as a way prevent overfitting to the identity context and enhance intra-class diversity.

- Face embeddings extracted using a **SFace-KT model**, pretrained on uniquely privacy-friendly synthetic data generated using a class-conditional GAN [8].

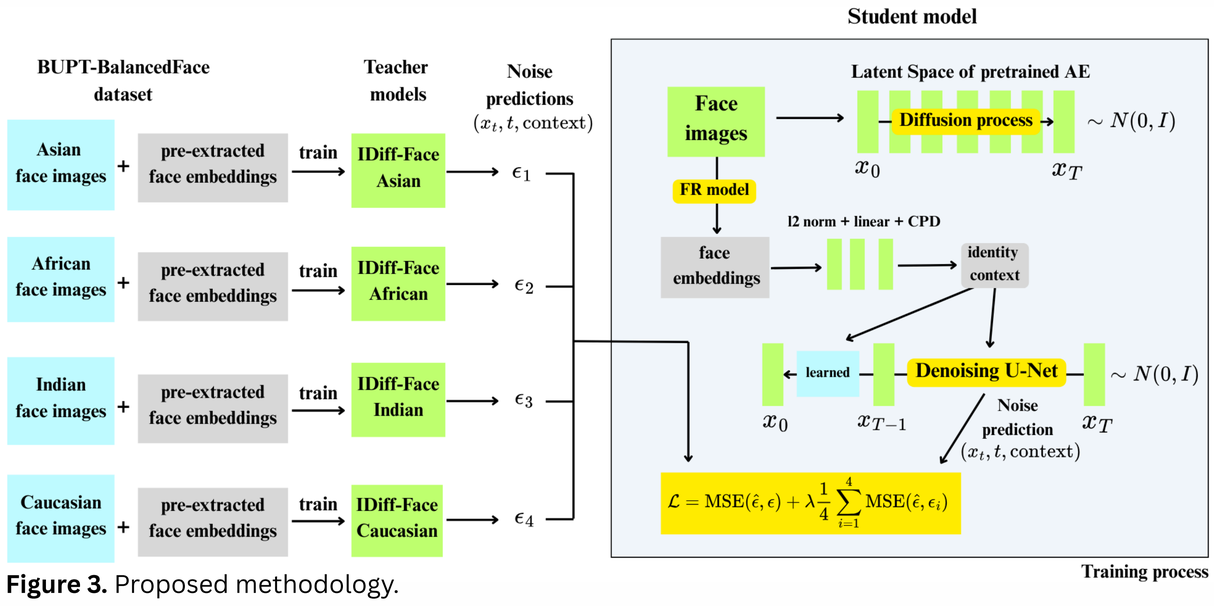

Figure 3. Proposed methodology.

SYNTHETIC FACE IMAGE PHOTOREALISTIC QUALITY

- Evaluate using pixel-based metrics such as the **Structural Similarity Index Measure (SSIM)**, and its multiscale version, **MS-SSIM**, as well as model-based metrics such as the **Inception Score (IS)**, the **Fréchet Inception Distance (FID)** and the **Learned Perceptual Image Patch Similarity (LPIPS)**.
- Use the Racial Faces in the Wild (RFW) [9] as the real dataset.
- **Baseline** for our results: synthetic dataset replaced with the BUPT-BalancedFace dataset.

SYNTHETIC FACE IMAGE UTILITY

- Evaluate by comparing the **accuracy of FR models** trained on our data with one standard baseline.

3. PRELIMINARY RESULTS

- **SAMPLING**

Figure 4. IDiff-Face: CPD 25% + KD . Uniform sampling. 380 identities, each with 50 variations (19000 images total)

- **PHOTOREALISTIC QUALITY**

	SSIM	MS-SSIM	FID	IS	LPIPS
Synthetic	**0.2914**	**0.2535**	**31.37**	1.063	**0.4139**
Baseline	0.2410	0.1570	31.69	**1.082**	0.4637

Table 1. Photorealistic quality metrics. We used a subset of 2560 images of each dataset.

4. CONCLUSIONS

- The preliminary results show that samples generated using our approach are comparable to real data in terms of photorealistic quality.

FUTURE WORK

- Compute the photorealistic quality metrics in a larger scale; refine their computation.
- Assess the utility of the generated samples for training FR.
- Experiment with different losses for KD.

5. ACKNOWLEDGEMENTS

This work was developed within the Component 5 - Capitalization and Business Innovation, integrated in the Resilience Dimension of the Recovery and Resilience Plan within the scope of the Recovery and Resilience Mechanism (MRR) of the European Union (EU), framed in the Next Generation EU, for the period 2021 - 2026, within project NewSpacePortugal, with reference 11.

REFERENCES

[1] Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, L 119, 1–88.

[2] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act). Official Journal of the European Union, L, -

[3] F. Boutros, V. Struc, J. Fierrez, and N. Damer, "Synthetic data for face recognition: Current state and future prospects," Image and Vision Computing, vol. 135, p.104688, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0262885623000628

[4] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," in Proceedings of the 32nd International Conference on Machine Learning, ser. Proceedings of MachineLearning Research, vol. 37. PMLR, 2015, pp. 2256–2265 [Online]. Available: http://proceedings.mlr.press/v37/sohl-dickstein15.html

[5] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, vol. 2022-June, pp. 10 674–10 685, 12 2021. [Online]. Available: https://arxiv.org/abs/2112.10752v2

[6] M. Wang and W. Deng, "Mitigating bias in face recognition using skewness-aware reinforcement learning," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 9322–9331.

[7] F. Boutros, J. H. Grebe, A. Kuijper, and N. Damer, "IDiff-Face: Synthetic-based face recognition through fizzy identity-conditioned diffusion models," Proceedings of the IEEE International Conference on Computer Vision, pp. 19 593–19 604, 2023.

[8] F. Boutros, M. Huber, P. Siebke, T. Rieber, and N. Damer, "Sface: Privacy-friendly and accurate face recognition using synthetic data," 2022 IEEE International Joint Conference on Biometrics, IJCB 2022, 6 2022. [Online]. Available: https://arxiv.org/abs/2206.10520v1

[9] Wang, M., Deng, W., Hu, J., Tao, X., & Huang, Y. (2019). Racial faces in the wild: Reducing racial bias by information maximization adaptation network. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 692–702).